

5 (b). Compute $\int_2^5 \frac{1}{(t+1)^2} dt$.

Sol. Always, when you meet $\int$, recall that the antiderivative $F(x)$ of $f(x)$ satisfies $F'(x) = f(x)$. We start from changing $\frac{1}{(t+1)^2}$ to $(t+1)^{-2}$.

Now look for an antiderivative of $(t+1)^{-2}$. Recall the rule $\int (ax+b)^n dx = \frac{1}{n+1} \frac{1}{a} (ax+b)^{n+1} + C$ when $n \neq -1$. This is exactly that situation where your $a = 1, b = 1$, and $n = -2$. Noting that our variable is not x but t , we get

$$\int (t+1)^{-2} dt = \frac{1}{-2+1} \frac{1}{1} (t+1)^{-1} + C = -(t+1)^{-1} + C.$$

You can check that the derivative of $-(t+1)^{-1} + C$ for any constant C is $(t+1)^{-2}$.

When calculating definite integral (this means you have numbers attached to the integral), you can set the integration constant C to be 0. Then we get $-(t+1)^{-1}$ as an antiderivative. Now plug in those numbers.

$$\begin{aligned} \int_2^5 \frac{1}{(t+1)^2} dt &= [-(t+1)^{-1}]_2^5 \\ &= -(5+1)^{-1} - (-(2+1)^{-1}) \\ &= -\frac{1}{6} + \frac{1}{3} = \frac{1}{6} \end{aligned}$$

6 (a). Evaluate the following: $\int (\sqrt{x} + 8x^3 + \frac{1}{e^{5x}} + \frac{1}{x+2} + 8) dx$.

Sol. Recall the rule we used in 5(b) to integrate $(ax + b)^n$ and that $\int \frac{1}{ax+b} dx = \frac{1}{a} \ln |ax + b| + C$. Also, $\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$.

Change all five items you have to integrate so that they fit into these forms. $\sqrt{x} = x^{\frac{1}{2}}$, and $\frac{1}{e^{5x}} = e^{-5x}$. We will integrate each term (connected by +) one by one and combine them.

Then we can use the same rule as in 5(b) but with $a = 1, b = 0, n = \frac{1}{2}$ for $\sqrt{x}$.

$$\begin{aligned}\int \sqrt{x} dx &= \int x^{\frac{1}{2}} dx = \frac{1}{\frac{1}{2} + 1} \frac{1}{1} x^{\frac{1}{2} + 1} + C \\ &= \frac{1}{\frac{3}{2}} x^{\frac{3}{2}} + C = \frac{2}{3} x^{\frac{3}{2}} + C\end{aligned}$$

For the remaining terms,

$$\int 8x^3 dx = 2x^4 + C.$$

$$\int \frac{1}{e^{5x}} dx = \int e^{-5x} dx = \frac{1}{-5} e^{-5x} + C$$

$$\int \frac{1}{x+2} dx = \ln |x+2| + C$$

$$\int 8 dx = 8x + C.$$

Now combine.

$$\int \left(\sqrt{x} + 8x^3 + \frac{1}{e^{5x}} + \frac{1}{x+2} + 8 \right) dx = \frac{2}{3} x^{\frac{3}{2}} + 2x^4 - \frac{1}{5} e^{-5x} + \ln |x+2| + 8x + C.$$

- 6 (b). Find the area enclosed by the curves $y = x^3 + 4$ and $y = x^3 + x^2$ between their points of intersection.

Sol. Area between 2 curves. Start with finding the points of intersection. These points are where the two graphs meet, so you set the graphs equal to each other. Here, $x^3 + 4 = x^3 + x^2$, so $4 = x^2$. Then $x = 2$ or $x = -2$.

Next, figure out which graph is on top of the other graph. You can do this by picking one x value in the middle of two intersection points and comparing y values of each graph there. I will pick 0 in between 2 and -2 . Plug in $x = 0$ in the first graph. This gives $y = 4$. Now plug in $x = 0$ in the second graph. You get $y = 0$. 4 is bigger than 0. This means the first graph is on top of the second graph in the middle of the two intersections.

Now set up area integral by $\int_{\text{smaller } x\text{value of intersections}}^{\text{the larger one}} \text{top graph} - \text{bottom graph } dx$. In this problem,

$$\begin{aligned} \int_{-2}^2 x^3 + 4 - (x^3 + x^2) dx &= \int_{-2}^2 4 - x^2 dx \\ &= \left[4x - \frac{1}{3}x^3 \right]_{-2}^2 \\ &= \left(8 - \frac{8}{3} \right) - \left(-8 - \frac{(-8)}{3} \right) \\ &= 8 + 8 - \frac{8}{3} - \frac{8}{3} = \frac{32}{3} \end{aligned}$$

- 7 (a). For the function $f(x, y) = 3x + 4xy + e^{xy}$, find $\frac{\partial f}{\partial y}$ and $\frac{\partial^2 f}{\partial x \partial y}$.

Sol. Recall that when getting partial in one variable, the other variable is treated like a constant. This means $\frac{\partial(3x)}{\partial y} = 0$. Also, just like $\frac{\partial(e^{2y})}{\partial y} = 2e^{2y}$, we have $\frac{\partial(e^{xy})}{\partial y} = xe^{xy}$. Thus,

$$\frac{\partial f}{\partial y} = 0 + 4x + xe^{xy}.$$

For the other part,

$$\begin{aligned} \frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (4x + xe^{xy}) \\ &= 4 + e^{xy} + xye^{xy}. \end{aligned}$$

Notice that the product rule $((uv)' = u'v + uv')$ was used in taking partial on xe^{xy} .

- 7 (b). For the function $f(x, y) = x^2 - y^3 + 6x + 12y + 1$, use the First Derivative Test to find the (x, y) coordinates of all the points at which the function possibly has a relative maximum or minimum.

Sol. This means we have to find the critical points. Just like before, you take derivative (in this case, f is in two variable, so partial derivatives) of the given f , and set them equal to 0.

$$\frac{\partial f}{\partial x} = 2x + 6 = 0$$

gives $x = -3$ and

$$\frac{\partial f}{\partial y} = -3y^2 + 12 = 0$$

gives $y = 2$ and $y = -2$. Now combine them in all possible ways. The critical points are $(-3, 2)$ and $(-3, -2)$. The problem wanted candidates and nothing else, so we can stop here. The answer will be $(-3, 2)$ and $(-3, -2)$.

- 7 (c). Use

$$D(x, y) = \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2$$

to determine whether the point $(-3, 2)$ you got in (b) is the location of a relative max, min, or neither (saddle point).

Sol. This continues from 7(b). To get the second partials in $D(x, y)$, we will use the first partials we computed in (b).

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (2x + 6) = 2,$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (-3y^2 + 12) = -6y,$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \frac{\partial f}{\partial y} = \frac{\partial}{\partial x} (-3y^2 + 12) = 0.$$

We are ready to compute $D(x, y)$.

$$D(x, y) = 2(-6y) - (0)^2 = -12y.$$

At the point $(-3, 2)$, our $x = -3$ and $y = 2$, so $D(-3, 2) = -12(2) = -24 < 0$. Negative D value at the point indicates that the point is a saddle point. Thus, $(-3, 2)$ is a saddle point.